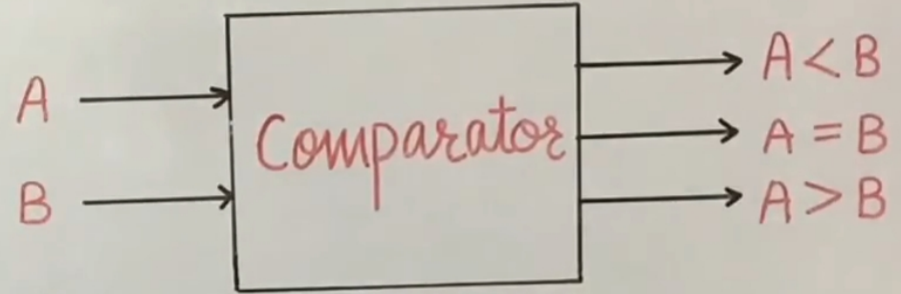


2-BIT COMPARATOR

	A		B				
	A ₁	A ₀	B ₁	B ₀	A < B	A = B	A > B
A = 0	0	0	0	0	0	1	0
	0	0	0	1	1	0	0
	0	0	1	0	1	0	0
	0	0	1	1	1	0	0
A = 1	0	1	0	0	0	0	1
	0	1	0	1	0	1	0
	0	1	1	0	1	0	0
	0	1	1	1	1	0	0
A = 2	1	0	0	0	0	0	1
	1	0	0	1	0	1	0
	1	0	1	0	1	0	0
	1	0	1	1	1	0	0
A = 3	1	1	0	0	0	0	1
	1	1	0	1	0	1	0
	1	1	1	0	1	0	0
	1	1	1	1	1	0	0

A digital Comparator is a combinational circuit designed to compare two n-bit no.



$$\begin{aligned}
 2^n &\rightarrow \text{no. of bits} \\
 2^2 &= 4 \\
 2^{2 \times 2} &= 2^4 = 16
 \end{aligned}$$

2-BIT COMPARATOR

A		B				
A ₁	A ₀	B ₁	B ₀	A < B	A = B	A > B
A=0	0	0	0	0	1	0
	0	0	1	0	0	0
	0	1	0	0	0	0
	0	1	1	0	0	0
	1	0	0	1	0	0
A=1	1	0	1	1	0	0
	1	1	0	0	0	1
	1	1	1	0	0	0
	1	1	1	0	0	0

$$(A < B) = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0$$

A ₁ A ₀	00	01	11	10
B ₁ B ₀				
00				
01	1			
11	1	1		1
10	1	1		

$$(A = B) = \bar{A}_1 \bar{A}_0 \bar{B}_1 \bar{B}_0 + \bar{A}_1 \bar{A}_0 \bar{B}_1 B_0 + A_1 A_0 B_1 B_0 + A_1 \bar{A}_0 B_1 \bar{B}_0$$

$$= \bar{A}_1 \bar{B}_1 (\bar{A}_0 \bar{B}_0 + A_0 B_0)$$

A ₁ A ₀	00	01	11	10
B ₁ B ₀				
00	1			
01		1		
11			1	
10				1

$$(A = B) = (A_0 \odot B_0) (A_1 \odot B_1) + A_1 B_1 (\bar{A}_0 B_0 + \bar{A}_0 \bar{B}_0)$$

$$= \bar{A}_1 \bar{B}_1 (A_0 \odot B_0) + A_1 B_1 (\bar{A}_0 \odot B_0)$$

$$= (A_0 \odot B_0) (\bar{A}_1 \bar{B}_1 + A_1 B_1)$$

$$= (A_0 \odot B_0) (A_1 \odot B_1)$$

A ₁ A ₀	00	01	11	10
B ₁ B ₀				
00		1	1	1
01			1	1
11				
10			1	

$$(A > B) = A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0$$

2-BIT COMPARATOR

A		B				
A ₁	A ₀	B ₁	B ₀	A < B	A = B	A > B
0	0	0	0	0	1	0
0	0	0	1	0	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	0	0
0	1	1	0	0	0	0
0	1	1	1	0	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	0	1	1	0	0	0
1	1	0	0	0	0	0
1	1	0	1	0	0	0
1	1	1	0	0	0	0
1	1	1	1	0	1	0

$$(A < B) = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0$$

A₁A₀ / B₁B₀

	00	01	11	10
00				
01	1			
11	1	1		1
10	1	1		

$$(A = B) = (A_0 \oplus B_0) (A_1 \oplus B_1)$$

A₁A₀ / B₁B₀

	00	01	11	10
00	1			
01		1		
11			1	
10				1

$$A > B =$$

A₁A₀ / B₁B₀

	00	01	11	10
00		1	1	1
01			1	1
11				
10			1	

$$(A > B) = A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0$$

